Roll Number: _____ Name: _____

Thapar Institute of Engineering & Technology, Patiala

Department of Computer Science and Engineering

Mid Sem Examination

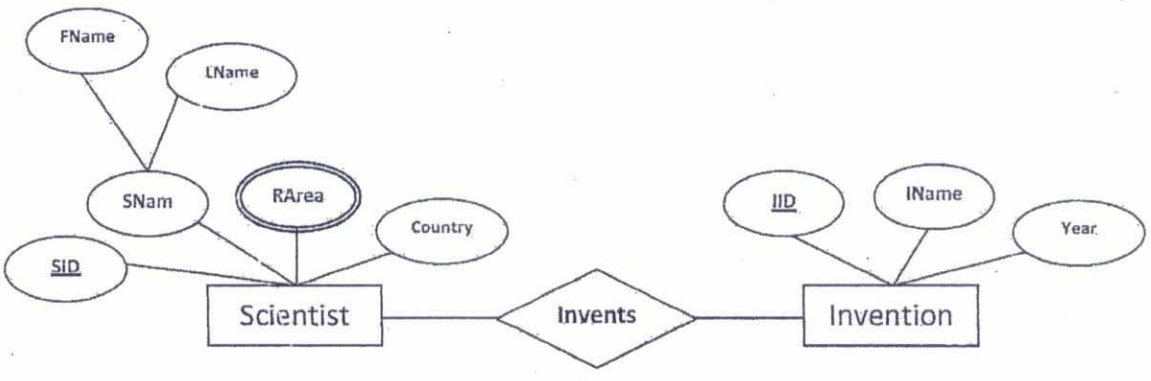
Max. Marks: 30 | **Course Name: Database Management System**

Course Code: UCS310

Time: 02 Hours | **B.E- Summer Sem**

Date: 02 July 2025, 8:00 AM to 10:00 AM | **Name of Faculty: Dr. Vibha Jain**

Note: ***Assume missing data, if any, suitably and mention the same. Ignore extra spaces, typing mistakes etc. **All the answer should be attempted sequentially.***

Q.No	Questions	Marks
Q1	a) A college stores student records in separate files: one for personal details, another for marks, and a third for library activity. Explain how data redundancy and inconsistency might arise in such a system. b) Explain the following with example: i. 3-schema architecture of a DBMS. ii. Integrity rule of a DBMS	3 3
Q2	a) Draw an ER diagram for the following scenario: A university has departments. Each department offers multiple courses. A course is taught by one or more professors. Students enroll in courses. Identify entities, relationships, attributes, and constraints clearly and show the cardinality for each connectivity from one entity to another entity. b) Convert the ER diagram into relational schemas with minimum number of tables. Specify the primary key and foreign key attributes in each relational schema.	3 3
Q3	a) Consider the relational database given below, where the primary keys are underlined. Give an SQL to express each of the following queries: Worker(WID, Name, Salary, Role) Factory(Dept_ID, DName, WID, budget) i. Obtain the total number of employees working in each department ii. List employee ID, for those with salary greater than max budget amongst all departments iii. List the names workers who work in departments having a budget greater than 1,00,000.	2*3
Q4	a) Discuss the differences between the following terms with examples: i. Referential Integrity and Entity Integrity constraint. ii. Primary key, candidate key, super key, and alternate key with proper examples. iii. Inner, Natural and Outer Join. iv. Generalization and Specialization	2*4
Q5	a) Convert the given E-R diagram into Tables. Explicitly mention primary key and foreign key for each of the table. 	4

********** All the best **********